



Savindu Suranjith

Computer Science & Engineering UG

Contact



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[Savindu Thennakoon](#)



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Tech Stack

Programming Languages

Python, TypeScript, JavaScript, C++, SQL

Frontend & Mobile

React, Next.js, React Native, Tailwind CSS, HTML5 Canvas

Backend & Architecture

FastAPI, Node.js, Express, Microservices, RabbitMQ

Databases & Cloud

PostgreSQL, Supabase, MongoDB, MySQL, Firebase

AI & Computer Vision

LangChain, LangGraph, MediaPipe, TensorFlow.js, RAG Pipelines, Multi-Agent Systems

DevOps & Tools

Docker, Git, Postman, Linux

Profile

Computer Science and Engineering undergraduate with hands-on experience in full-stack web development, cross-platform mobile engineering, AI agent orchestration, and computer vision systems. Proven background in deploying real-time pose tracking frameworks, responsive client architectures, and interactive digital tools. Experienced with JavaScript/TypeScript, Node.js, Express, React Native, Supabase, MERN stack, TensorFlow.js, and MediaPipe. Interested in building intelligent, motion-driven mobile applications and modern web platforms that bridge AI models with intuitive user experiences.

Education

University of Moratuwa

2024 - Present

B.Sc. (Hons) in Engineering

Computer Science and Engineering

Dharmapala college Bandarawela

2014 - 2023

- G.C.E Advanced Level Examination (2023)
Physical Science Stream
(3As - Zscore - 2.12)

- G.C.E. Ordinary Level Examination (9As)

(2019)

Projects

Ongoing

Smart Attendance and Classroom Access Platform- [GitHub Repository](#) (Group project)

- Tech Stack: React Native, React, Next.js, Supabase, Tailwind CSS, Face Recognition, GPS Geofencing**
- Platform Architecture: Built an enterprise-grade academic attendance platform with a microservices architecture using FastAPI, Next.js, React Native, PostgreSQL, and RabbitMQ.
- Edge AI & Anti-Spoofing: Designed an on-device verification pipeline leveraging MiniFASNet, facial landmark tracking, and 192D embeddings to detect presentation attacks in real time.
- Fraud Prevention: Combined biometric checks with GPS geofencing and randomized verification windows to eliminate proxy attendance.
- Privacy-by-Design: Computed vector embeddings directly on-device, preventing raw facial imagery or biometric templates from leaving the client.
- Concurrency & Scalability: Offloaded vector matching to distributed RabbitMQ message queues for asynchronous processing during high-volume check-in bursts.

2026 Feb - ongoing

FitLock (AI App-Locker) | [View on Google Play](#)

- Tech Stack: React Native, React, Vite, Capacitor, MediaPipe Pose, On-Device ML**
- Developed FitLock, a cross-platform digital wellbeing application that locks target apps behind physical exercise requirements (squats, push-ups, jumping jacks).
- On-Device Computer Vision: Integrated MediaPipe Pose via HTML5 Canvas for client-side, zero-latency repetition counting and joint angle calculation without streaming or storing user video.
- Monetization & Privacy Compliance: Integrated Google AdMob alongside the Google User Messaging Platform (UMP) SDK to handle monetization with full GDPR consent compliance.

Interests

- Android App Development
- Software Architecture
- Web Development
- Software Development
- Linux
- NLP
- AI & ML
- Game Development
- Computer Vision

Languages

- English
- Sinhala

Soft Skills

- Collaboration
- Time Management
- Leadership
- Teamwork
- Ability to adapt quickly
- Problem-Solving

References

- **Prof. Dulani Meedeniya**
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- **Dr. Adeesha Wijayasiri**
Senior Lecturer Department of
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May 2026
HackerRank Orchestrate May - AI Agent | [GitHub Repository](#)

- Tech Stack: Python, LangGraph, Google GenAI (Gemma/Gemini), ChromaDB, HuggingFace all-MiniLM-L6-v2, Pydantic, Asyncio**
- Deterministic State Machine: Built a multi-node LangGraph workflow orchestrating classification, safety checks, policy routing, and ticket resolution across HackerRank, Claude, and Visa domains.
 - Local RAG Pipeline: Indexed multi-domain support corpora into an embedded ChromaDB using HuggingFace all-MiniLM-L6-v2 embeddings, ensuring grounded, hallucination-free retrieval without external web dependencies.
 - Type-Safe Routing & Structured Output: Utilized Google Generative AI paired with Pydantic schema models (.with_structured_output) to enforce strict CSV classification and triage decision compliance.
 - Resilience & Concurrency: Designed an asynchronous processing engine using asyncio.Semaphore and staggered scheduling alongside Tenacity exponential retry policies to handle high ticket volume and transient API rate limits.
 - Ranked 81st out of 1400+ submissions (12000 + registrants)

June 2026
PrajaNavigator (AI Insight Platform) | [GitHub Repository](#)

- **Tech Stack: Python, RAG, Computer Vision, React, Tailwind CSS, Docker**
- System Overview: Engineered a full-stack, multi-agent public administration platform to streamline citizen services via multimodal OCR verification, pre-counter readiness scoring, and automated form generation.
- Hierarchical RAG Architecture: Designed a 3-layer Hierarchical Retrieval-Augmented Generation (RAG) engine incorporating automated conflict resolution and multi-agent workflow orchestration with offline fallback modes.

Certificates

- Professional Agentic Architect Certification (ongoing) | Google cloud
- Google play store listing certificate | Google play academy

Leadership & Extracurricular Activities

- Leadership Roles
- Marketing Committee Lead | MazeX 1.0 - Micromouse competition (2026)
 - MATLAB webinar series - IET | Co-chair (July 2026)
- Societies
- IEEE RAS | Web and Marketing pillar member (Feb 2026 – Present)
 - Moraspirit Marketing pillar member (March 2025 – Present)
 - Operational pillar Member | IET On Campus (March 2026 – Present)
- Sports
- Team Member | Taekwondo, University of Moratuwa (2025)